JAMES DE RAJA

Senior Real-Time Performance Engineer · Unity Rendering · XR · Online Multiplayer

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Senior Unity Performance Engineer with 13+ years optimizing real-time games and large-scale systems. Improved performance **4× at 10K entities** and optimized systems handling **5M+ executions/day**. Architecting an open-world shooter with 10K+ entities, multiplayer systems, and XR rendering pipeline. See detailed case studies at james.alphaden.club

Core Strengths: Real-time Performance Optimization · Large-Scale Simulation · XR Rendering

TOP RESULTS

- **40 ms → 9.4 ms · 4× frame-time reduction** at 10K entities under heavy load (ECS architecture, ~24 → ~106 FPS, no visual change). View Case Study
- Identified critical GPU bottleneck in XR rendering (+7.27 ms overdraw cost — 201 transparent draws, stereo 90 Hz). View Case Study
- **66 → 136 FPS** · 10K-mesh GPU instancing; CPU render-thread collapsed, render no longer CPU-bound. View Lab

CURRENT FOCUS — OPEN-WORLD SHOOTER

Update Strategies at Scale — ECS Architecture · View Case Study · View Repo

- **4× frame-time reduction (40 → 9.4 ms; 24 → 106 FPS)** at high-density enemy scenes under sustained simulation load — no visual change.
- Benchmarked OOP, manager, and ECS architectures under load to identify the only strategy that scales past 5K entities without frame collapse.

XR Stress Lab — Unity 6 / URP / OpenXR · View Case Study · View Repo

- **+7.27 ms GPU delta isolated in stereo XR rendering** — 201 transparent draws at 90 Hz; GPU cost doubled when overdraw compounded with MSAA 4×. Findings now define the shooter's XR rendering strategy.
- Built targeted diagnostic tooling (overdraw analysis, MSAA cost isolation, GPU timing overlays) to surface bottlenecks before they hit the production pipeline.

GPU Instancing Lab · View Lab

- **66 → 136 FPS (15.04 → 7.34 ms)** at 10K identical meshes; CPU render-thread collapsed, now the default rendering path for enemy archetypes.

PRODUCTION GAME WORK

Sneaky Warrior 3D — Frame-Budget Stabilization · View Case Study · App Store

- **Held 16 ms across mobile tiers including tile-based GPUs** with 100+ animated enemies, ragdolls, and projectile bursts.
- Optimized large-scale enemy rendering using batching and spatial grouping while preserving culling efficiency.
- Off-screen skinning gated; performance spikes profiled and mitigated. Patterns now applied in the open-world shooter.

8-Player Real-Time Multiplayer — 25K+ Downloads

- Built real-time multiplayer system delivering stable 60 FPS across 8 players under sustained input-saturation stress.
- Designed multiplayer architecture supporting 8 concurrent players under full input-saturation load and adversarial network conditions — now the foundation layer for the open-world shooter.

Supersonic Hypercasual Competition — Round 1 Winner ($2M prize pool)

- Selected from a global pool judged on CPI, D1 retention, and FTUE; shipped through Supersonic's live-ops pipeline.
- Repeat publisher relationships with Voodoo, Lion Studios, and Supersonic across 50+ commissioned prototypes and 100+ public releases. View Portfolio

DISTRIBUTED SYSTEMS ENGINEERING

Member Leadership Staff (Staff Engineer) · Zoho Corporation · Jun 2017 – Present · Chennai

9 years building high-scale distributed systems with strict uptime and reliability requirements.

- **Led performance-focused system redesigns** enabling 99.9% uptime and zero-downtime migrations across high-scale infrastructure (5M+ daily executions).
- **Owned and scaled Zoho CRM integration** powering 5M+ daily executions across 8,000+ organizations. Led large-scale system migration with zero customer disruption.
- Built 5 of the top 10 most-used integrations on Zoho Flow; 100+ integrations shipped overall.
- Architected a **zero-downtime environment-variable system for Zoho Creator** — 1M+ executions/day across 1,500+ orgs, zero customer-reported incidents post-release.
- **Led remediation of a production server-crash incident**: reduced system load 70–85% with API contract unchanged (3M executions/day, 2,500+ orgs).
- Mentor 6+ active mentees, including 2 direct reports; total mentorship footprint of 20+ engineers across Zoho product teams.

TECHNICAL SKILLS

Rendering & Performance: Unity (URP, Built-in), ECS / DOTS, GPU Instancing, Frame Debugger, GPU Timing Analysis, RenderDoc

XR & Systems: OpenXR, Unity XR Toolkit, Stereo Rendering Optimization, real-time performance optimization

Profiling: Unity Profiler, RenderDoc, GPU Timing Queries, custom runtime overlay

Backend: Java, Distributed REST APIs / Webhooks, High-scale systems (5M+ executions/day), Git, GitHub Actions

EDUCATION

B.E., Electronics & Communication — SSN College of Engineering · 2013–2017